WORKED SOLUTION

STEP III 2009, Question 11

Mechanics • Variable Mass Motion

STEP III 2009, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q11

The key idea

When mass changes, $F = ma$ is the wrong law: we must use force = rate of change of momentum. The dust is stationary, so it brings no momentum of its own, and the comet's momentum $M(1+bx)v$ changes only because of the external force. That gives a first integral straight away. Writing $v = \mathrm{d}x/\mathrm{d}t$ then turns it into a separable equation whose solution is a quadratic in x.

Solution

We model the comet as a particle. The dust it collects was at rest, so it carries no momentum into the comet, and the only force on the system (comet plus collected dust) is the gravitational force Mf along the direction of motion.

Part (i)

With $f = 0$ there is no external force, so momentum is conserved. At entry the momentum is MV; at time t it is $M(1+bx)v$. Hence

$$(1+bx)\,v = V.$$

Now put $v = \dfrac{\mathrm{d}x}{\mathrm{d}t}$ and separate:

$$\int_0^x (1+bu)\,\mathrm{d}u = \int_0^t V\,\mathrm{d}s \quad\Longrightarrow\quad x + \tfrac{1}{2}bx^2 = Vt,$$

using $x = 0$ when $t = 0$. So $bx^2 + 2x - 2Vt = 0$, and since $x \geq 0$ we need the root with the positive square root:

$$x = \frac{\sqrt{1+2bVt} - 1}{b}.$$

Part (ii)

Now the external force is Mf, so

$$\frac{\mathrm{d}}{\mathrm{d}t}[M(1+bx)v] = Mf.$$

Since M and f are constants, integrating from 0 to t gives $M(1+bx)v - MV = Mft$ (at $t=0$, $x=0$ and $v=V$). Therefore

$$v = \frac{ft+V}{1+bx},$$

as required.

Ask yourself Why can't we just write $M(1+bx)\dfrac{\mathrm{d}v}{\mathrm{d}t} = Mf$?

That equation would ignore the momentum needed to bring each new speck of dust from rest up to speed v. The product rule shows the difference: the rate of change of momentum is $M(1+bx)\dot{v} + Mb\dot{x}\,v$, and the second term is exactly that "catching-up" cost.

As in part (i), write $v = \mathrm{d}x/\mathrm{d}t$ and separate:

$$\int_0^x (1+bu)\,\mathrm{d}u = \int_0^t (fs+V)\,\mathrm{d}s \quad\Longrightarrow\quad x + \tfrac{1}{2}bx^2 = \tfrac{1}{2}ft^2 + Vt.$$

So $bx^2 + 2x - (ft^2 + 2Vt) = 0$ and, taking the non-negative root again,

$$x = \frac{\sqrt{1+bft^2+2bVt}-1}{b}.$$

(Setting $f=0$ recovers part (i).)

Constant speed. The comet moves with constant speed V exactly when $x = Vt$ for all t, that is, when $\sqrt{1+bft^2+2bVt} = 1+bVt$. Squaring, this needs

$$1 + bft^2 + 2bVt = 1 + 2bVt + b^2V^2t^2 \quad \text{for all } t,$$

which holds precisely when $f = bV^2$. Then $v = \dfrac{bV^2t+V}{1+bVt} = V$, so with this choice the comet does move at constant speed: the gravitational pull exactly supplies the momentum needed to accelerate the dust it sweeps up.

Limiting speed. In general, using $1+bx = \sqrt{1+bft^2+2bVt}$ from the formula for x,

$$v = \frac{ft+V}{\sqrt{1+2bVt+bft^2}} = \frac{f+V/t}{\sqrt{bf+2bV/t+1/t^2}} \longrightarrow \frac{f}{\sqrt{bf}} = \sqrt{\frac{f}{b}} \quad \text{as } t\to\infty,$$

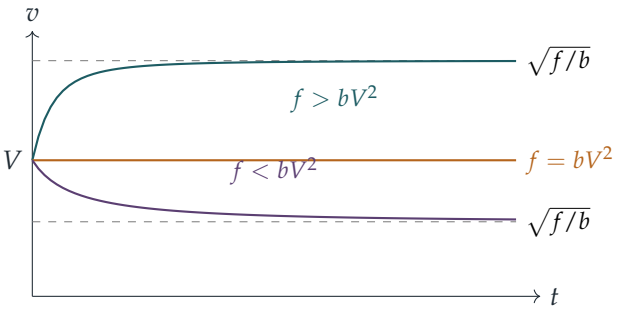
where we divided top and bottom by $t>0$ and used $bf>0$ (we need $f>0$, since the force acts in the direction of motion). So the speed tends to the constant $\sqrt{f/b}$.

Ask yourself Is this consistent with the constant-speed case?

Yes: if $f = bV^2$ then $\sqrt{f/b} = V$, the speed the comet had all along. In fact differentiating v^2 gives

$$\frac{\mathrm{d}(v^2)}{\mathrm{d}t} = \frac{2(ft + V)(f - bV^2)}{(1 + 2bVt + bft^2)^2},$$

so the speed rises steadily towards $\sqrt{f/b}$ if $f > bV^2$, falls steadily towards it if $f < bV^2$, and stays at V if $f = bV^2$.



Checks

- Units: b has units of $1/\text{length}$ and f is an acceleration, so f/b has units of speed squared and $\sqrt{f/b}$ is a speed; bft^2 and bVt are dimensionless, as they must be inside the square root.
- With $b = 1$, $V = 1$, $f = 3$, stepping the equations $\dot{x} = p/(1 + bx)$, $\dot{p} = f$ numerically agrees with the formulas for x and v to eight decimal places, and at $t = 10^6$ the speed is $1.7320508\ldots \approx \sqrt{3}$.
- As $b \to 0$ (no dust), $x = \left(\sqrt{1 + bft^2 + 2bVt} - 1\right)/b \to \frac{1}{2}ft^2 + Vt$, the usual constant-acceleration result.

Common slips

- Using $(\text{mass}) \times \dot{v} = \text{force}$ for a body whose mass is changing.
- Taking the negative root of the quadratic, which gives $x < 0$.
- Forgetting the constant of integration MV when integrating the momentum equation.

Beyond STEP

Variable-mass problems are a classic first-year topic: the same momentum bookkeeping gives the rocket equation, and raindrops growing as they fall through a cloud lead to

equations very like these. The limiting speed $\sqrt{f/b}$ is a terminal velocity, where the force exactly balances the rate at which momentum is spent on new mass. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats systems of variable mass carefully.

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